

Prediction Task	Best Baseline Model	Best Baseline AUROC (95% CI)	Best Fusion Strategy	Best Fusion AUROC (95% CI)	Absolute Change
In-Hospital Mortality	Numerical	0.9010 (0.8891-0.9131)	No Synergy Observed	0.9100 (0.8968-0.9204)	+0.0090
30-Day Readmission	Semantic	0.6618 (0.6328-0.6926)	No Synergy Observed	0.6712 (0.6412-0.7020)	+0.0094
Vasopressor Intervention	Numerical	0.7719 (0.7394-0.8035)	Neural Network	0.7862 (0.7548-0.8176)*	+0.0143
Mechanical Ventilation	Numerical	0.6939 (0.6602-0.7271)	Hierarchical	0.7023 (0.6690-0.7356)	+0.0084
Length-of-Stay > 3 Days	Numerical	0.7272 (0.7146-0.7401)	Dynamic Weight	0.7324 (0.7198-0.7452)	+0.0052
Length-of-Stay > 7 Days	Numerical	0.7438 (0.7203-0.7682)	Ensemble	0.7499 (0.7268-0.7747)	+0.0061